

Effective Numerical for Capturing Main Trait in the Rare Mutation Limit

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Abstract

Key words.

1 Introduction

{section1}

Understanding how populations evolve under the joint effects of selection, mutation, and competition is a central question in evolutionary biology and mathematical ecology. At the microscopic level, stochastic individual-based models represent evolution as the outcome of random birth, death, and mutation events. In the large population limit, these stochastic models can be rigorously connected to deterministic integro-differential or partial differential equations describing the dynamics of population densities [?]. In our paper, we consider a trait-structured population model, which has proven to be powerful tools for studying adaptation and selection. In this setting, each individual carries a phenotypical trait $\theta \in \mathbb{R}$ that determines its fitness. And we use $n(t, x, \theta)$ to describe the population density at time t , position x and with the trait θ .

2 Summary

{section6}

In this work, we

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